

Functionality specification

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< Disaster Response and Managment System development >

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Name	Role	Company	Date	Signature

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Introduction

+ The State of Arstotzka has announced a tender for the development of a Disaster Response and Management System (DRMS) to enhance emergency preparedness, improve response coordination, and minimize the impact of natural and man-made disasters. The DRMS is designed to be accessible and reliable under extreme conditions.

The DRMS integrates a web-based command center application, a mobile app for on-ground responders, and an IoT system connected to sensors placed in disaster-prone areas. These sensors monitor seismic activity, flood levels, and air quality, transmitting real-time alerts to the system.

Emergency responders use the mobile app to receive alerts, update their current status, and report incidents as they occur. The app includes geolocation features to provide navigation to affected areas and allows **responders** to upload photos, videos, and notes about disaster conditions. **Responders** can also communicate directly with **the command center** through in-app messaging, ensuring seamless coordination. This communication is performed through priority channels provided by telecom providers, following the regulation of using reserved channels.

The web application serves as the central hub for disaster management administrators. It provides tools to visualize real-time data from IoT sensors and field reports through dashboards and heatmaps. Administrators can deploy resources such as rescue teams, medical supplies, and equipment to affected areas, tracking their progress live. The system also generates reports to analyze the effectiveness of response efforts post-disaster.

The DRMS is designed to be accessible and reliable under extreme conditions. Both applications must work offline, syncing data with the server once connectivity is restored. The mobile app will support English, Spanish, Arabic, and Hindi to cater to diverse teams, while the web app will be available in English and Spanish. Communication between IoT devices, mobile apps, and the web system will be encrypted, ensuring data security.

Scope of the project

+ The DRMS shall include a **web-based command center application for administrators and a mobile app for on-ground responders, connected to an IoT sensor network**, allowing them to communicate, cooperate, gather data, analyze, hence enhance performance in real time.

To elaborate, **the mobile app** built for onsite emergency responders allows them to receive alerts, update their status, and report incidents as they occur. It enables live photo, video and note uploading about the disaster conditions. More so, it incorporates in-app direct messaging with the command center, important in the need of additional assistance. The web-based command center application provides administrators with real-time analysis of data gathered from the IoT system and mobile application.

While in the case of network loss, the application system must locally store and synchronize data post regaining regular connection, it **will not include** any resources necessary to ensure its active work offline,

nor the priority channels provided by telecom providers, following the regulation of using reserved channels.

Concepts

+ **IoT system of sensors** implies using a network of sensors that collect information, in this case seismic activity, flood levels or air quality, updating the system in real time geolocating the measurement and provide visualization of collected data for administrators.

+ **Data synchronisation** implies that in the case of network connection fail among parties, the data should be stored locally and updated as soon as the reconnection allows it. This is crucial for the post disaster analysis.

Role description

+ **Disaster management administrators** analyze data sent in by the IoT sensors and on-site reporters. Additionally, they should be able to **communicate with emergency responders**, deploy more help and resources on the disaster site and track their statuses in real time. Post disaster, it is the management's duty to provide a follow up report.

+ **On-ground emergency responders** communicate with disaster management administrators through receiving and reporting live alerts. They send photos, videos, and notes about disaster conditions. Additionally, they communicate with disaster management administrators via messaging.

Assumptions and dependencies

+ Third party geolocation, as the features on the mobile app depend on the GPS satellite data.

+ Telecom providers, as they need to grant and maintain access to reserved priority channels for emergency communication.

+ Hardware support, as the functionality of the monitoring system depends on the continuous operation and maintenance of the IoT sensor system.

List of requirements

- + admin receives information from the IoT sensor system as the trigger
- + admin sends immediate notification to the responders, especially in proximity
- + responders update their status live
- + admin contacts the responders

Requirements related to functionality

Admin communicates with responders

An **administrator** must be able to contact and communicate with the responder on site to evaluate if additional help is needed, and to declare the disaster ending. The web app allows direct communication

with the responders via in-app messaging, receiving live updates and real time notes, photos, and video uploads.

Responder messages the admin

Responders must be able to contact the central hub and can do so via an in-app messaging feature.

Responder sends live updates

When the **responder** is on site, he ought to collect additional information for data analysis. The mobile app features a step that allows one to upload live photos or videos, note updates.

Admin deploys help

Upon analyzing the situation, the **administrator** must send help accordingly. The web app must allow one to send an immediate notification to all the responders, especially the ones in proximity to the disaster.

Admin analyzes collected data

To ensure enough help is on the way, and to enhance future performance, the **administrator** must analyze information live, as well as post the disaster. During the disaster, the software creates visualization out of the information sourced through the IoT, and displays it through the web app, helping their understanding of the vastness of the situation. The data stores, and after the disaster, can be used for determining the efficiency of the previous rescue.

Requirements related to characteristics

User language selection

As a user, both a **responder and/or an admin**, I want to be able to choose the language I am most fluent in, as it ensures that all the information I acquire or provide is clear, and the focus is on solving the task, rather than possible delays due to lack of understatement. The web app supports English and Spanish, and the mobile app will support English, Spanish, Arabic and Hindi to cater to diverse teams.

Data synchronization

As a user, **both responder and especially an admin**, I do not want to miss any important data from the other end due to brief network breaks. The app should gather and synchronize post reconnection; both the mobile and the web application ought to have this feature.

Security data encryption

As a user, **both responder and especially an admin**, I do not want any information being tampered with. To prevent this, I want communication among all parties to be encrypted, ensuring data security.